

# Yin-Chung (Lloyd) Huang

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## EDUCATION

### University of Sydney (USYD)

School of Life and Environmental Science, Faculty of Science

PhD Candidate

Oct. 2023 – Present

Sydney, Australia

- Advisor: Prof. Budiman Minasny
- Fully funded by Postgraduate Research Scholarship in Soil Monitoring

### National Taiwan University (NTU)

Master of Science in Agricultural Chemistry

Sept. 2020 – June 2022

Taipei, Taiwan

- Advisor: Prof. Zeng-Yei Hseu
- Thesis: A feasibility study of using pXRF and Vis-NIR reflectance spectra as the diagnostic criteria of podzolic soils in subalpine areas of Taiwan

### National Taiwan University

Bachelor of Science in Agricultural Chemistry

Sept. 2016 – June 2020

Taipei, Taiwan

- Honors: **Summa Cum Laude** (Rank 1/34)
- Biological Statistics Program (focus: advanced statistics, applied linear algebra, artificial neural networks, and spatial analysis)

## PUBLICATIONS

### First-author refereed journals

**Huang, Y. C.**, Padarian, J., Ng, W., Minasny, B., & McBratney, A. B. (2026). Zero-shot inference with Tabular Prior-data Fitted Network (TabPFN) for soil MIR spectral analysis. *Geoderma*, 471, 117880.

**Huang, Y. C.**, Ng, W., Minasny, B., Tang, Y., & McBratney, A. B. (2025). Accessible soil spectroscopy: Evaluating low-cost Vis-NIR spectrometers for resource-constrained environments. *European Journal of Soil Science*, 76, e70248.

**Huang, Y. C.**, Padarian, J., Minasny, B., & McBratney, A. B. (2025). Using Monte Carlo conformal prediction to evaluate the uncertainty of deep learning soil spectral models. *SOIL*, 11, 553-563.

**Huang, Y. C.**, Ng, W., Minasny, B., & McBratney, A. B. (2025). Characterising and quantifying soil clay-sized minerals using mid-infrared spectroscopy. *Soil & Tillage Research*, 252, 106590.

**Huang, Y. C.**, Huang, C. Y., Minasny, B., Chen, Z. S., & Hseu, Z. Y. (2023). Using pXRF and Vis-NIR for characterizing diagnostic horizons of fine-textured podzolic soils in subtropical forests. *Geoderma*, 437, 116582.

**Huang, Y. C.**, Chen, Z. S., Hsu, J. Y., Chiu, C. M., & Hseu, Z. Y. (2022). Differentiation of fine-textured podzolic soils controlled by climate and landscape in Taiwan. *Geoderma*, 428, 116155.

**Huang, Y. C.**, & Hseu, Z. Y. (2021). Silicon availability in relation to soil properties in Inceptisols on uncultivated lands and paddy fields in Taiwan. *Geoderma Regional*, 26, e00406.

### Co-author refereed journals

Tang, Y., Tran, T. K. A., Minasny, B., Bakhshandeh, S., Du, M., Francos, N., **Huang, Y. C.**, Jang, H. J., Ng, W., Xue, P., & McBratney, A. B. (2026). SOC stabilisation shifts from carbon accumulation in temperate soils to mineral association in subtropical soils. *Soil Biology and Biochemistry*, 215, 110101.

Swan, T., Jang, H. J., **Huang, Y. C.**, Fidelis, C., Yinil, D., Bala, B., Das, B. S., & Field, D. (2025). Comparative analysis of Vis-NIR and MIR spectroscopy for predicting soil properties and identifying minerals at smallholder cocoa farms across Papua New Guinea. *Soil Advances*, 5, 100094.

Minasny, B., Bandai, T., Ghezzehei, T. A., **Huang, Y. C.**, Ma, Y., McBratney, A. B., Ng, W., Norouzi, S., Padarian, J., Rudyanto, Shariffar, A., Styc, Q., & Widyastuti, M. (2024). Soil Science-Informed Machine Learning. *Geoderma*, 452, 117094.

### Manuscripts under review

**Huang, Y. C.**, Ng, W., O'Donoghue, T., Minasny, B. & McBratney, A. B. Field portable soil spectroscopy: What's beyond Vis-NIR? *Computers and Electronics in Agriculture*.

### International conference publications

**Huang, Y. C.**, Padarian, J., Minasny, B., & McBratney, A. B. (2025). Evaluating the uncertainty of deep learning soil spectral models using Monte Carlo conformal prediction (MC-CP). Abstract Proceedings

- of EuroSoil 2025. European Confederation of Soil Science Societies. Seville, Spain. Sep. 8-12, 2025. (oral presentation)
- Huang, Y. C.**, Ng, W., Minasny, B., & McBratney, A. B. (2024). Characterizing soil clay minerals using mid-infrared spectroscopy. Abstract Proceedings of Centennial of the IUSS. International Union of Soil Science. Florence, Italy. May 19-21, 2024. (oral presentation)
- Huang, Y. C.**, Huang, C. Y., Minasny, B., Chen, Z. S., & Hseu, Z. Y. (2023). Characterizing diagnostic horizons of podzolic soils in Taiwan using pXRF and Vis-NIR. Abstract Proceedings of 4<sup>th</sup> Global Soil Security Conference. Korean Society of Soil Science and Fertilizer. Seoul, Korea. June 26-29, 2023. (oral presentation)
- Huang, Y. C.**, Huang, C. Y., Chen, Z. S., & Hseu, Z. Y. (2023). Using pXRF and Vis-NIR as an interpretable model to predict soil properties in podzolic soils of subtropical forest. Abstract Proceedings of 2<sup>nd</sup> joint Workshop of the IUSS Working Groups Digital Soil Mapping and Global Soil Map. International Union of Soil Science. Orléans, France. Feb. 7-9, 2023. (oral presentation)
- Huang, Y. C.**, Huang, C. Y., & Hseu, Z. Y. (2022). Characterization of podzolic soils using digital soil morphometrics in subtropical subalpine forests. Abstract Proceedings of 22<sup>nd</sup> World Congress of Soil Science. International Union of Soil Science. Glasgow, UK. July 31-Aug. 5, 2022. (oral presentation)
- Huang, Y. C.**, Huang, C. Y., & Hseu, Z. Y. (2022). Characterization of organometallic complexes in podzolic soils in subalpine forest using proximal sensors. Abstract Proceedings of 8<sup>th</sup> International Symposium on Soil Organic Matter. Korean Society of Soil Science and Fertilizer. Seoul, Korea. June 26-30, 2022. (oral presentation by pre-recorded video)
- Huang, Y. C.**, & Hseu, Z. Y. (2021). Characterization of podzolic soils using digital morphometrics in a subtropical subalpine forest of Taiwan. Abstract Proceedings of Pedometric Webinar. International Union of Soil Science. June 16-17, 2021. (oral presentation)
- Huang, Y. C.**, & Hseu, Z. Y. (2019). Silicon availability and its relation to other properties of soils in Taiwan. Abstract Proceedings of 14<sup>th</sup> International Conference of the East and Southeast Asia Federation of Soil Science Societies (ESAFS). National Taiwan University. Taipei, Taiwan. Nov. 3-8, 2019. (oral presentation)
- Huang, Y. C.**, Tsui, C. C., Chiang, C. H., & Hseu, Z. Y. (2018). Element composition of the colluvial soils at an archeological site from Linkou Terrace, Taiwan. Abstract Proceedings of Joint Seminar on Environmental Ecology and Restoration. Ishikawa Prefectural University. Hakodate, Japan. Sept. 3-6, 2018. (oral presentation)

#### *Domestic conference publications*

- Huang, Y. C.**, Padarian, J., Minasny, B. & Syu, C. H. (2025). Using convolutional neural network for digital soil mapping in Taiwan. Abstract Proceedings of Annual Meeting of the Chinese Society of Soil and Fertilizer Sciences. National Chung Hsing University. Taichung, Taiwan. Nov. 24, 2025. (poster presentation)
- Huang, Y. C.**, Huang, C. Y., & Hseu, Z. Y. (2022). Using pXRF and Vis-NIR spectroscopy to predict soil organic carbon. Abstract Proceedings of Annual Meeting of the Chinese Society of Soil and Fertilizer Sciences. National Taiwan University. Taipei, Taiwan. Dec. 8, 2022. (poster presentation; **first place**)
- Huang, Y. C.**, Huang, C. Y., & Hseu, Z. Y. (2022). Using dry chemistry to quantify soil properties. Abstract Proceedings of Annual Meeting of the Chinese Environmental Analytical Society. National Central University. Taoyuan, Taiwan. Aug. 31-Sept. 1, 2022. (oral presentation)
- Huang, Y. C.**, & Hseu, Z. Y. (2021). Characterization of podzolic soils in Shanlinshi using digital soil morphometrics. Proceedings of Annual Meeting of the Geography Society of China located in Taipei. National Changhua University of Education. Webinar. May. 29, 2021. (oral presentation; **honorable mention**)
- Huang, Y. C.**, Huang, C. Y., & Hseu, Z. Y. (2020). Combining Vis-NIR spectroradiometer and pXRF to study pedogenesis. Abstract Proceedings of Annual Meeting of the Chinese Society of Soil and Fertilizer Sciences. National Chung Hsing University. Taichung, Taiwan. Dec. 10, 2020. (poster presentation; **honorable mention**)

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#### PROFESSIONAL SERVICE

##### **Reviewer Service**

- Soil Science Society of America Journal
- Geoderma
- Soil Use and Management

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#### RESEARCH EXPERIENCE

**Soil Survey and Remediation Laboratory, NTU**  
*Research Assistant*

Aug. 2022 – Aug. 2023

- Lab PI: Prof. Zeng-Yei Hseu
- Established spectral library for Taiwanese soils and built a partial least squared regression model to predict the organic carbon content in soils
- Combined portable X-ray fluorescence spectrometer and Vis-NIR spectroradiometers to predict soil organic carbon, clay, amorphous iron, and optical density of oxalate extractant in podzolic soils
- Investigated the differentiation of podzolic soils in subalpine forests in Taiwan using selective extraction, micromorphology, and multivariate statistics

## AWARDS & HONORS

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Travel Grants supported by International Union of Soil Sciences</b><br><i>1000 EUR as reimbursements of travel costs for early-career scientists</i> | May 2024              |
| <b>First Place for Poster Competition</b><br><i>Annual Meeting of the Chinese Society of Soil and Fertilizer Sciences</i>                               | Dec. 2022             |
| <b>Travel Grants supported by International Union of Soil Sciences</b><br><i>1000 USD as reimbursements of travel costs for early-career scientists</i> | July 2022             |
| <b>Honorable Mention for Academic Thesis Competition</b><br><i>Annual Meeting of the Geography Society of China located in Taipei</i>                   | May 2021              |
| <b>Honorable Mention for Poster Competition</b><br><i>Annual Meeting of the Chinese Society of Soil and Fertilizer Sciences</i>                         | Nov. 2020             |
| <b>Presidential Award (Academic Excellence Award)</b><br><i>Top 5% of the class in a semester</i>                                                       | Oct. 2020             |
| - Demonstrated diligence on every subject                                                                                                               | Oct. & Apr. 2019      |
| - Completed all credits with outstanding grades                                                                                                         | Oct. & Apr. 2018      |
|                                                                                                                                                         | Oct. & Apr. 2017      |
| <b>Hsin Tian Kong Long-term Scholarship</b><br><i>1500 USD each semester</i>                                                                            | July 2019 – June 2022 |
| - Acquired financial support from one of the most famous charity groups in Taiwan                                                                       |                       |
| - Long-term mentorship program for young scientists                                                                                                     |                       |

## TEACHING EXPERIENCE

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|-------------------------------------------------------------------------------------------------|------------------------|
| <b>USYD SOIL2005 Soil and Water: Earth's Life Support Systems</b><br><i>Demonstrator</i>        | Feb. 2025 – June 2025  |
| - Assisted teaching in lab work section                                                         |                        |
| <b>USYD ENVX1002 Introduction to Statistical Methods</b><br><i>Demonstrator</i>                 | Feb. 2024 – June 2024  |
| - Delivered lectures on R programming and statistics                                            |                        |
| <b>USYD ENVX4001 GIS, Remote Sensing and Land Management</b><br><i>Demonstrator &amp; Tutor</i> | Feb. 2024 – June 2024  |
| - Assisted teaching, project design and final presentation                                      |                        |
| <b>NTU Soil Science Lab (2020, 2021)</b><br><i>Teaching Assistant</i>                           | Sept. 2021 – Jan. 2022 |
| - Organized weekly soil science lab lecture                                                     | Sept. 2020 – Jan. 2021 |